Supplement S3: External protocol and budget results to "What Does a Persona Add?"

Evidence lineage, paired capacity comparison, and external stimulus protocol

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1 Scope and source lineage

This supplement records the retrospective B1 capacity comparison and R2 stimulus-matched protocol and results. The historical diagnostics are supplied separately as Supplement S4. Protocol freezing is internal, not public preregistration.

The external source is `LLM-Digital-Twin/Twin-2K-500`. Its pinned revision is `f883165a3026fde855dfd448e0cd16443ab257b6`

The original 49-endpoint catalog protocol is `TWIN_PROTOCOL_FROZEN.json`. Its file SHA-256 is `0ea4652877ccc93b288e71172fb5a4d473985c431b794fb095434746622fae17`

The respondent-specific R2 amendment is `TWIN_STIMULUS_AMENDMENT.json`. Its SHA-256 is `1a968951bbf7f11a5d9a838b00676e058de1896a7778bece04ba0b1d3eb70067`

It binds the parent protocol, prepared receipt, question-only sidecar, FIT orientation audit, canonical target-count ledger, and presentation-slot response semantics. The source and FIT-only receipts are the `source audit` and `FIT orientation audit`. The R2 amendment was frozen *after* the earlier v6 history-only CAL readback but *before* any R2 fit or GPU forecast. All 84 compute cluster source tests and complete R2 input validation are engineering checks, not predictive results.

2 B1: complete paired capacity comparison

The target is normalized ordinal ranked probability score (RPS) on 60 disjoint target items for each of the same 600 E3 people. For K categories,

$$\mathrm{RPS}(p, y) = \frac{1}{K-1} \sum_{k=1}^{K-1} \left(\sum_{j=1}^{k} p_j - \mathbf{1}\{y \le k\} \right)^2 .$$

Lower is better. The contrast is the archived E3 frozen hybrid minus the within-budget, cross-validated statistical predictor. The hybrid's per-person loss is averaged across nine fixed correlated combinations (three old LLM seeds times three ridge subset seeds); the reference's per-person loss is averaged across three fixed subset seeds. Forecast probabilities are scored before averaging losses. The nine historical combinations are not nine independent optimizer runs. The 600 respondent differences, not individual item answers, are the inferential units.

At each budget $n \in \{80, 160, 320, 1000, 8000, 8320\}$ and in each of the raw-answer and exact categorical-profile representations, the frozen candidate families were an empirical prior, ridge penalties $\{1, 10, 100, 1000\}$, and a small nonlinear kNN/ExtraTrees family. Within each n and each of three nested subset seeds, three-fold training-only cross-validation selected family and hyperparameters by mean held-person RPS, with registered ties. The $8,320$ pool comprises the historical Train8000, Search80, Select120 and Calibrate120 people; no E3 Eval600 label entered fitting or selection. All 18 budget-by-seed selections and 108 family forecasts were retained. The historical hybrid itself nevertheless used additional search, prior, selection and calibration supervision, so n is a *statistical fit budget*, not total unique human information or literal questionnaire answers saved.

Table 1: All 12 retained B1 contrasts. Δ is hybrid minus selected reference normalized RPS; positive favors the reference. Intervals resample 600 respondents conditional on fixed forecasts. $p_{H,12}$ is the separate 12-contrast secondary Holm family; the two $n = 8320$ rows have primary Holm-adjusted $p_{H,2} = 0.0004$.

n	History	Δ	Conditional 95% CI	$p_{H,12}$
80	Raw	−0.007086	[−0.007705, −0.006471]	0.0024
160	Raw	−0.000500	[−0.001047, 0.000066]	0.0820
320	Raw	0.005355	[0.004644, 0.006076]	0.0024
1000	Raw	0.012844	[0.011920, 0.013785]	0.0024
8000	Raw	0.021580	[0.020406, 0.022741]	0.0024
8320	Raw	0.021679	[0.020556, 0.022857]	0.0024
80	Categorical	−0.008770	[−0.009444, −0.008102]	0.0024
160	Categorical	−0.002250	[−0.002824, −0.001697]	0.0024
320	Categorical	0.004270	[0.003631, 0.004925]	0.0024
1000	Categorical	0.011452	[0.010559, 0.012307]	0.0024
8000	Categorical	0.018254	[0.017158, 0.019373]	0.0024
8320	Categorical	0.018311	[0.017188, 0.019426]	0.0024

The receipt records 5,000 respondent bootstrap draws and 5,000 paired two-sided sign flips. Both $n = 8320$ primary contrasts have Holm-adjusted $p = 0.0004$, but neither passes the *benefit* gate: $\Delta \leq -0.001$, upper CI below -0.001, and Holm $p < 0.05$ were all required together. At $n = 80$ the hybrid leads despite its hidden historical supervision; $n = 160$ differs by representation; from $n = 320$ both selected references lead. These are conditional estimates on the already exposed E3 panel.

3 R2 respondent-specific stimulus crosswalk

The original catalog stores the first question it saw for a QID, while the survey presented 40 product categories in randomized slots and with respondent-specific prices. A question-only source scan read the respondent ID and source question objects, not the Wave-4 numeric answer column. For each respondent it masked exactly one posted price, matched the remaining text to a canonical product, preserved that product's actual price and ordered options, and checked that every product appeared once. The nine nonpricing target texts and options were invariant. The full ledger is:

Verified item	Count
Respondents / canonical endpoints	2,058 / 49
Product offers / products per respondent	82,320 / 40
Product-slot pairs observed	1,600
Price levels per product / zero-price offers	11 / 7,490
Offers exactly matching old catalog example	220
Nonpricing endpoints with invariant text/options	9
FIT-only question-text matches	43,000 / 43,000
FIT-only numeric slot-choice agreements	49,000 / 49,000

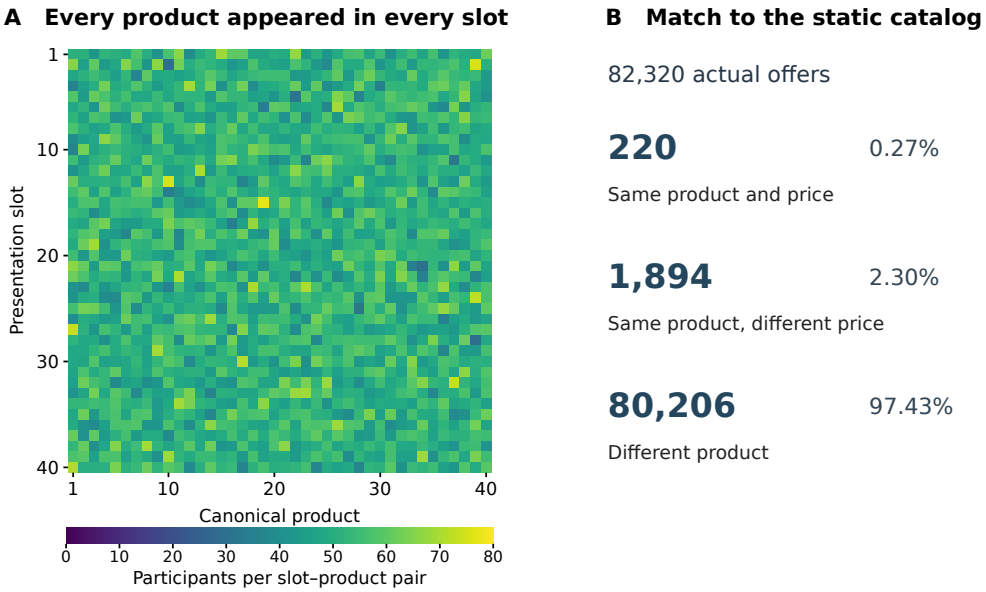


Figure 1: Outcome-blind product-by-presentation-slot counts (left) and comparison with the static catalog example (right). Of 82,320 actual offers, 220 matched both catalog product and price, 1,894 matched product but not price, and 80,206 displayed a different product in the catalog's slot. Counts derive from question text and price only; no participant answer enters the figure.

The numeric orientation audit inspected only the 1,000 frozen FIT people to establish that `QID9_j` numeric response columns index randomized *presentation slots*. Its 49,000 agreements connect selected option positions in the source question object to the numeric export. DEV, CAL and TEST answers were not inspected for this orientation check; no predictive model was fit or scored by it. R2 maps a raw slot answer to a canonical product answer using the verified question-only slot/product permutation. Both LLM arms receive the exact respondent question and both statistical controls receive the same product/price/condition fields. The sidecar, amendment, option order, counts and source code are hash-bound at input load and comparison. The earlier v6 history-only CAL result (prior chosen for 45 of 49 endpoints) marginalizes over this offer variation and does not measure whether history adds value conditional on a known offer.

4 External endpoints, scoring, and primary aggregation

The frozen endpoint definition is in `TWIN_PROTOCOL_FROZEN.json`; the following table fixes its groups and scales without disclosing participant data.

Family	Canonical columns	K	Primary score
Economic preferences	`QID9_1`-`QID9_40`, 40 canonical products	2	Exact log loss
Heuristics/cognitive bias	`QID288_1-4`, `QID289_1-4`, eight ratings	7	Normalized RPS
Heuristics/cognitive bias	`QID196`, one binary choice	2	Exact log loss

All probabilities must match every respondent, endpoint, ordered option and split. Missing human labels use one fixed observed mask per endpoint across all methods; missing *forecasts* fail the panel. For nominal/binary endpoints, $L(p, y) = -\log p_y$ without clipping; $p_y = 0$ yields infinite loss and unavailable aggregate inference. Brier is secondary. For the eight ordinal tasks, L is the normalized RPS above with $K = 7$.

The frozen primary overall estimand is dimensionless. Let D_e be the mean CAL primary loss of the FIT1000-only Laplace prior with $\alpha = 1$ for endpoint e (zero history). For a system S and reference R, let

$$\delta_e = \frac{1}{|O_e|} \sum_{i \in O_e} \frac{L_{ie}(S) - L_{ie}(R)}{D_e}, \qquad \Delta = \frac{1}{2}\left(\frac{1}{40} \sum_{e \in E_{40}} \delta_e\right) + \frac{1}{2}\left(\frac{1}{9} \sum_{e \in H_9} \delta_e\right).$$

Here O_e is the fixed set of observed TEST labels at endpoint e. The freeze records each prior vector, FIT and CAL labeled counts, and D_e; $D_e \leq 10^{-6}$ or no CAL labels aborts. Every endpoint needs at least 30 observed TEST labels

or the *entire* primary aggregate is unavailable. This averages normalized *differences* within endpoint, then endpoints within family, then the two families equally; it never pools raw log loss and RPS. The three-subfamily variant was explicitly not chosen as primary. A respondent bootstrap or sign vector is shared across all 49 endpoints, retaining within-person dependence. Empty-mask bootstrap draws are counted and the valid-draw conditional CI and failure fraction reported. Holm correction covers the complete 12 primary system-versus-full1400 contrasts separately from the two T0-versus-neutral contrasts; endpoint tables use a separate secondary Holm family.

5 Frozen model, comparator, and failure gates

The R2 statistical family includes an empirical Laplace prior and predeclared regularized logistic and ExtraTrees candidates using history, respondent-specific stimulus, or both. Three-fold CV inside the 1,200 FIT+DEV people is specified to choose each endpoint's family and settings before CAL or TEST scoring. The `decoder1200` CAL forecast must be held out. The `full1400` comparator is specified to refit using FIT+DEV+CAL and to be scored only on TEST; its in-sample CAL output is ineligible for mixture tuning.

The two fixed LLM arms are neutral and the portable T0 instruction component, on each pinned model. The instruction is not the full old NEO genotype. A CPU preflight binds source hashes, model/tokenizer commits, architecture, exact complete prompt tokenization and a no-truncation context ceiling. A finite one-GPU DEV probe tests 10 people across every endpoint and both arms before CAL. Each forecast is the exact full-softmax mass on registered canonical digit tokens, normalized within that set; the frozen minimum digit-mass gate must pass. CAL alone is designated to choose, per arm and endpoint, $w \in \{0, 0.1, \ldots, 1\}$ for $wP_{\mathrm{LLM}} + (1-w)P_{1200}$ by the endpoint's primary score. The fixed $w = 0.5$ mixture is also retained. TEST starts only after the mixture freeze binds the R2 stimulus lineage, prepared source, options, and same model/readout. Direct arms, fixed mixtures and selected mixtures are all compared to full1400 without outcome-based deletion.

An incomplete forecast grid, mismatched labels or options, source-hash drift, failed digit-mass/context gate, nonfinite probability or unbound receipt is a failed or unavailable comparison, never an imputed zero effect. Exact-zero true-answer log loss remains infinite; no clipping or selective method dropout repairs it. Scores retain negative, mixed and failed rows. One model's engineering failure does not certify a scientific effect for the other. TEST remains sealed until the declared gates pass.

6 Resource and exposure ledger

Finite Slurm allocations are required for CPU prepare/selection/scoring and for single-GPU model inference. Receipts distinguish allocated CPU-hours, allocated GPU-hours, elapsed seconds, prompt tokens, attempted/completed forecasts, failed tasks, and unknown fields; an absent field is *unknown*, not zero. Both DEV models produced 980 forecasts; their combined GPU allocation was 1.69 hours. The pilot-based resource forecast for CAL and TEST was 218.52 GPU-hours across both models. The two CAL and two TEST GPU allocations used 143.89 hours in total. The historical minimum of 300.60 allocated GPU-hours excludes known-unsummed search/development and failure components, so it is not an end-to-end cost ratio against the fitted statistical controls. The external Wave-4 labels are public, model pretraining exposure is unknown, and the instrument measures hypothetical answers rather than purchases.

AI assistance is disclosed in the main manuscript.

7 Verified external TEST results

The separate numerical audit reconstructed all 14 primary and 686 endpoint contrasts, including the frozen respondent bootstrap and sign-flip draws. Negative values favor the system. The two contrasts against neutral form their own Holm family.

System	Reference	Δ	95% interval	Holm p
Granite Neutral	Full1400	0.43267	[0.41554, 0.44938]	0.0024
Granite Neutral half	Full1400	0.09614	[0.08774, 0.10418]	0.0024
Granite Neutral CAL	Full1400	-0.00404	[-0.00723, -0.00096]	0.0360
Granite T0	Full1400	0.37564	[0.35879, 0.39205]	0.0024
Granite T0 half	Full1400	0.08115	[0.07308, 0.08924]	0.0024
Granite T0 CAL	Full1400	-0.00384	[-0.00692, -0.00085]	0.0486
Granite T0	Granite neutral	-0.05703	[-0.06204, -0.05193]	0.0004
Qwen Neutral	Full1400	0.30872	[0.29367, 0.32358]	0.0024
Qwen Neutral half	Full1400	0.06902	[0.06117, 0.07686]	0.0024
Qwen Neutral CAL	Full1400	-0.00202	[-0.00503, 0.00106]	0.4107
Qwen T0	Full1400	0.33939	[0.32222, 0.35712]	0.0024
Qwen T0 half	Full1400	0.07664	[0.06800, 0.08536]	0.0024
Qwen T0 CAL	Full1400	-0.00110	[-0.00427, 0.00211]	0.5007
Qwen T0	Qwen neutral	0.03066	[0.02269, 0.03899]	0.0004

On the 40 pricing products, the separate exploratory price-plus-history minus price-only contrast was +0.00420 mean log loss (paired respondent 95% interval $[-0.00015, 0.00852]$). All 26,320 paired cells were observed and finite.

Exploratory direct comparison of Granite mixtures. Both contrasts are T0 minus neutral on 658 people and all 49 endpoints, using the frozen R2 normalization and equal family weights. Intervals are paired respondent percentile bootstrap intervals (10,000 draws, seed 20261110). The CAL-selected contrast is +0.000197 ($[-0.000541, 0.000957]$). Weights differed on 18 endpoints; mean Granite weights were 0.1020 (neutral) and 0.0980 (T0). At a common weight of 0.5, the contrast is −0.014990 ($[-0.017155, -0.012732]$). These are post hoc contrasts outside the registered Holm families. The analysis read frozen forecasts.

Exploratory mixture decomposition. Each CAL-selected mixture is compared with its own decoder1200. The bridge is decoder1200 minus Full1400. All 658 respondents and 49 endpoints were reused. Descriptive paired respondent 95% intervals use 10,000 draws (seed 20261110), outside the registered Holm families. Negative values favor the system. Decoder1200 vs Full1400: −0.001150 ($[-0.004417, 0.002007]$). Granite neutral: −0.002886 ($[-0.004478, -0.001282]$). Granite T0: −0.002689 ($[-0.004239, -0.001123]$). Qwen neutral: −0.000872 ($[-0.003070, 0.001314]$). Qwen T0: +0.000050 ($[-0.001549, 0.001636]$).